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Moneyness

money is best described as an adjective, not a noun

the blog of JP Koning

Sunday, October 19, 2014

Fedcoin



Recent posts by [Adrian Hope Baille](#) and [Sina Motamedi](#) have got me thinking again about the idea of the Federal Reserve (or any other central bank for that matter) adopting bitcoin technology. [Here's](#) an older post of mine on the idea, although this post will take a different tack.

The bitcoin ethos enshrines the idea of a world free from the totalitarian control of central banks. So in exploring the idea of Fed-run bitcoin-style ledger, I realize that I run the risk of being cast as Darth Vader (or even "yikes" the Emperor) by bitcoin true believers. [So be it](#). While I do empathize with the bitcoin ideal—I support freedom in banking—I rank the importance of *bitcoin-as-product* above *bitcoin-as-philosophy*. And at the moment, bitcoin is *not* a great product. While bitcoin has many useful features, these are all overshadowed by the fact that its price is too damn volatile for it to be taken seriously as an exchange medium. This volatility arises because bitcoin lacks a fundamental value, or anchor, a point that I've written about [many times in the past](#). However, there is one way to fix the crypto volatility problem...

Enter Fedcoin

Setting up the apparatus would be very simple. The Fed would create a new blockchain called *Fedcoin*. Or it might create a Ripple style ledger by the same name. It doesn't matter which. There would be an important difference between Fedcoin and more traditional cryptoleaders. One user—the Fed—would get special authority to create and destroy ledger entries, or Fedcoin. (Sina Motamedi gives a more [technical explanation](#) for how this would work in the case of a blockchain-style ledger)

The Fed would use its special powers of creation and destruction to provide two-way physical convertibility between both of its existing liability types—paper money and electronic reserves—and Fedcoin at a rate of 1:1. The outcome of this rule would be that Fedcoin could only be created at the same time that an equivalent reserve or paper note was destroyed and, vice versa, Fedcoin could only be destroyed upon the creation of a new paper note or reserve entry.

So unlike bitcoin, the price of Fedcoin would be anchored. Should Fedcoin trade at a discount to dollar notes and reserves, people would convert Fedcoin into these alternatives until the arbitrage opportunity disappears, and vice versa if Fedcoin should trade at a premium.

As for the supply of Fedcoin, it would effectively be left free to vary *endogenously*, much like how the Fed currently let's the market determine the supply of Fed paper money. This flexibility stands in contrast to the fixed supply of bitcoin and other cryptocurrencies. The mechanism would work something like this. Should the public demand Fedcoin, they would have to bring paper dollars to the Fed to be converted into an equivalent number of new Fedcoin ledger entries, the notes officially removed from circulation and shredded. As for banks, if they wanted to accumulate an inventory of Fedcoin, they would exchange reserves for Fedcoin at a rate of 1:1, those reserves being deleted from Fed computers and the coins added to the Fedcoin ledger.

Symmetrically, unwanted Fedcoin would reflux to the central bank in return for either newly-created cash (in the case of the public) or reserves (in the case of banks), upon which the Fed would erase those coins from the ledger. The upshot is that the Fed would have no control over the quantity of Fedcoin—it would only passively create new coin according to the demands of the public.

Apart from that, Fedcoin would be similar in nature to most other cryptoleaders. All Fedcoin transactions would be announced to a distributed network of listening nodes for processing and verification. In other words, these nodes, and not the Fed, would be responsible for maintaining the integrity of the Fedcoin ledger.

Why implement Fedcoin?

The main reasons that the Fed would implement Fedcoin would be to provide the public with an innovative and cheap payments option, and to provide the taxpayer with tax savings.

The public would enjoy all the benefits of bitcoin including fast transaction speeds, cheap transaction costs,

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and the ability to transact almost anywhere and with almost anyone as long as all parties to a transaction had a smartphone and the right software. At the same time Fedcoin's stability would immediately differentiate it from bitcoin. No longer would users have to fear losing 50% of their purchasing power prior to making a transaction.

Fedcoin's distributed architecture would be both complementary and in many ways superior to *Fedwire*, a centralized system which currently provides for the transfer of Fed electronic reserves among banks. I won't bother getting into the specifics: see [this](#) old post.

By introducing Fedcoin, the Fed would also lower its costs. While I haven't done the calculations, I have little doubt that running a distributed cryptolledger is far cheaper than maintaining billions of paper notes in circulation. Paper currency involves all sorts of outlays including designing and printing notes, collecting, processing and storing them, as well as constantly defending the note issue against counterfeiters. A distributed ledger does all this at a fraction of the cost. As Fedcoin begins to displace cash, and I think that this would steadily happen over time due to its superiority over paper, the Fed's costs would fall and its profits rise to the benefit of the taxpayer.

Fedcoin would have no impact on monetary policy

Fed officials might balk at giving the idea a shot if they feared that adopting a Fed cryptolledger would impede the smooth functioning of Fed monetary policy. They needn't worry.

The Fed currently exercises control over the price level by varying the quantity of reserves and/or the interest paid on reserves. The existence of cash doesn't get in the way of this process, nor has it ever gotten in the way. Bringing in a third liability type, Fedcoin, the quantity of which is designed to fluctuate in the same way as cash, would likewise have no impact on monetary policy. The Fed would continue to lever the return on reserves in order to get a bite on prices while allowing the market to independently choose the quantity of Fedcoin and cash it wished to hold.

Well, almost none: Interest on Fedcoin and the zero lower bound

Ok, I sort of lied in the last paragraph. While it happens only rarely, there are times when cash does get in the way of monetary policy, and so would Fedcoin if it were implemented. If the Fed needs to reduce rates on reserves to negative levels in order to hit its price and employment targets, the existence of cash impedes the smooth slide below zero. With reserves yielding -2% and paper notes yielding 0%, reserves would quickly be converted en masse into cash until only the latter remains. At that point the Fed would have lost its ability to alter rates—cash doesn't pay interest nor can it be penalized—and would no longer be capable of exercising monetary policy. This is called the zero-lower bound, and it terrifies central bankers.

Fedcoin has the potential to alleviate the zero lower bound problem. Here's how.

As Fedcoin adoption grows among the public, cash would steadily be withdrawn. And while it might not shrink to nothing—the public might still choose to use some cash—at least the Fed would have a good case for entirely canceling [larger denominations](#) like the \$100 and \$50.

Consider also that it would be possible for interest to be paid on each Fedcoin (unlike bitcoin and cash), the rate to be determined by the Fed. And just as Fedcoin could earn positive interest, the Fed could also impose a negative rate penalty on Fedcoin. This would effectively solve the Fed's zero lower bound problem. After all, if the Fed wished to reduce the rate on reserves to -2 or -3% in order to deal with a crisis, and reserve owners began to bolt into Fedcoin so as to avoid the penalty, the Fed would be able to forestall this run by simultaneously reducing the interest rate on Fedcoin to -2 or -3%. Nor could reserve owners race into cash, with only low denomination and [expensive-to-store](#) \$5s and \$10s available.

So by implementing something like Fedcoin, the Fed could safely implement a negative interest rate monetary policy.

(Lastly, monetary policy nerds will notice that the displacement of non-interest yielding cash with interest-yielding Fedcoin is a tidy way to arrive at Milton Friedman's optimum quantity of money, or the [Friedman rule](#).)

The big losers: banks

Fedcoin has the potential to tear down the private banking system. Interest yielding Fedcoin would be able to do everything a bank deposit could do and more, and all this at a fraction of the cost. As the public shifted out of private bank deposits and into Fedcoin, banks would have to sell off their loan portfolios, the entire banking industry shrinking into irrelevance.

One way to prevent this from happening would be for the Fed to make an explicit announcement that any bank could be free to create its own competing copy of Fedcoin, say *Wells Fargo Coin*. Like the Fed, Wells Fargo would promise to offer two-way convertibility between its deposits/cash/Fedcoin and Wells Fargo Coin at a rate of 1:1 to ensure that the price of its new ledger entries were well-anchored. The bank could then implement features to compete with Fedcoin such as higher interest rates or complimentary financial services. Even as Wells Fargo's deposit base steadily shrunk due to technological obsolescence, its base of Wells Fargo Coin liabilities would rise in a compensatory manner.

The resulting lattice network of competing private bank crypto ledgers built on top of the Fedcoin ledger would work in a similar fashion to the current banking system. Wells Fargo would make loans in Wells Fargo Coin and take deposits of FedCoin as well as competing bankcoins, say *CitiCoin* or *Bank of America Coin*. Intra-bank cryptocoin payments would be cleared on the books of the Federal Reserve with reserves transfers over the Fedwire funds system, although Fedcoin might eventually take the place of Fedwire. A change in the value of Fedcoin or reserves due to a shift in monetary policy would be transmitted immediately into a change in the value of all private bankcoins by virtue of the convertibility of the latter into the former.

Nor would it be necessary to start with Fedcoin and then introduce bankcoins. Why not begin with the latter and skip Fedcoin altogether? Why aren't private banks at this very moment switching out deposits and replacing them with cryptolledgers?

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KYC: Know your customer

'Know your customer' regulations would make implementation difficult, but not impossible.

With bitcoin, the location of a coin (its address) is public but the identity of the owner is not. However, laws require banks to gather information on their customers to protect against money laundering. As these laws are unlikely to change with the advent of new technology, banks would probably require anyone wanting to use bank cryptolegders to have an account with a regulated bank. This would not be too onerous given that most Americans already have bank accounts. However, it compromises anonymity, one of the key ideals of bitcoin, since each coin would be traceable by the authorities to a real person.

Perhaps there is still a way to preserve some degree of anonymity. Historically the Fed has always been spared from KYC rules since it has never had to document who uses cash. By grandfathering KYC exemption to Fedcoin, any user who wanted to preserve their anonymity could use Fedcoin rather than any of the multiple bankcoin ledgers, just like today they prefer to use anonymous Fed cash rather than bank accounts to transact.

In summary

So that's a rough sketch of Fedcoin—a decentralized, flexible, and well-backed payments system that grants one user, the Fed, a set of special privileges and responsibilities. Feel free to modify the idea in the comments section.

And just so we are keeping tabs, these are the institutions that Fedcoin could eventually make obsolete: bank deposits, banks (unless the latter are allowed to innovate their own bankcoins), the credit card networks Visa and Mastercard, bank notes, Fedwire, and even bitcoin itself, which would be unable to compete with a stable-value copy of itself.

Bitcoin true believers may not like this post, but perhaps they can take something constructive from it. Fedcoin is one of the potential competitors in the distant horizon. Now is the time for the rebels to figure out how to create a stable-price version of bitcoin, before Darth Vader does it himself. Otherwise they may someday find themselves closing down their bitcoin startups in order to write code for the Empire.

Note: My apologies to readers for my having succumbed to the constant temptation to adorn all blog posts with Star Wars references.

Posted by [JP Koning](#) at [1:28 PM](#)

Labels: [anonymity](#), [bitcoin](#), [cash](#), [Fedcoin](#), [Federal Reserve](#), [Fedwire](#), [fintech](#), [Friedman rule](#), [Ripple](#), [zero lower bound](#)

25 comments:



Nick Rowe October 19, 2014 at 4:36 PM

Very good post JP. (I was composing a comment on the ZLB, and the technical possibility of + and - interest on Fedcoin, and then you answered it in the next paragraph.)

The implications of KYC was a totally new thought for me.

If Fedcoin does come into being, and private banks can't compete, the Fed's balance sheet would need to expand a lot. And if it moves towards the Friedman Rule, it would need to expand even more. It would need to own everything! That's gonna be the catch. Do we want central banks owning mortgages, and making private loans? It is similar to nationalising the whole banking system, and the whole of Finance. (My post on Fractional reserves and the optimum quantity of money.)

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Nick Rowe October 19, 2014 at 4:37 PM

"can" " should be "can't "



JP Koning October 19, 2014 at 7:57 PM

Glad you liked it, Nick.

"If Fedcoin does come into being, and private banks can't compete, the Fed's balance sheet would need to expand a lot. "

Yes, I agree.

Here's the link for Nick's post for the curious:

http://worthwhile.typepad.com/worthwhile_canadian_initi/2014/09/fractional-reserves-and-the-optimum-quantity-of-money.html

And no, I don't think we want the central bank to nationalize the banking system. Hopefully allowing the private banks to establish their own competing bankcoins might return some balance to the system. Perhaps there might be some other ways to avoid the nationalization scenario.

Jussi October 29, 2014 at 5:20 AM

Great post.

Banks would be normal intermediaries then. They would need to pay more interest than the Central Bank or seek other forms of funding, long term or equity, more aggressively. So should the interest paid on Fedcoin take that into account.

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